

✓ Q1. Let R1 (a, b, c) and R2 (x, y, z) be two relations in which a is the foreign key of R1 that refers to the primary key of R2 . Consider following four options. *1/1

- (a) Insert into R 1 (b) Insert into R 2 (c) Delete from R 1 (d) Delete from R 2

Which of the following is correct about the referential integrity constraint with respect to above?

- ☐ Operations (b) and (c) will cause violation.
- ☐ Operations (c) and (d) will cause violation.
- ☒ Operations (d) and (a) will cause violation. ✓
- ☐ Operations (a) and (b) will cause violation.

✓ Q2. Which of the following statements is FALSE about weak entity set? * 1/1

- ☐ Weak entity set avoids the data duplication and consequent possible inconsistencies caused by duplicating the key of the strong entity.
- ☐ Weak entities can be deleted automatically when their strong entity is deleted.
- ☐ A weak entity set has no primary keys unless attributes of the strong entity set on which it depends are included.
- ☒ Tuples in a weak entity set are not partitioned according to their relationship with tuples in a strong entity set. ✓

✓ Q3. In RDBMS, different classes of relations are created using _____ technique to prevent modification anomalies. *1/1

- ☐ Referential Integrity
- ☒ Normal Forms ✓
- ☐ Data Integrity
- ☐ Functional Dependencies

✗ Q4. BCNF is not used for cases where a relation has _____. * 0/1

- ☐ Two mutually exclusive foreign keys
- ☐ Two (or more) candidate keys
- ☒ Two candidate keys and composite ✗
- ☐ The candidate key overlap

Correct answer

- ☒ Two mutually exclusive foreign keys

✓ Q5. Database applications were built directly on top of file system to overcome the following drawbacks of using file-systems: (i) Data redundancy and inconsistency (ii) Difficulty in accessing Data (iii) Data isolation (iv) Integrity problems *1/1

- ☐ (i) only
- ☐ (ii) and (iv)
- ☒ (i) , (ii) , (iii) and (iv) ✓
- ☐ (i) , (ii) and (iii)

✓ Q6. In E-R model, Y is the dominant entity and X is subordinate entity then which is TRUE *1/1

- ☒ If Y is deleted, then X is also deleted ✓
- ☐ If Y is deleted, then X is not deleted.
- ☐ If X is deleted, then Y is also deleted.
- ☐ None of the above

✓ Q7. In functional dependency Armstrong inference rules refers to: * 1/1

- ☐ Reflexivity, Augmentation and Decomposition
- ☒ Transitivity, Augmentation and Reflexivity ✓
- ☐ Augmentation, Transitivity, Reflexivity and Decomposition
- ☐ Reflexivity, Transitivity and Decomposition



Thanks for filling out this form: DBMS Quiz 3

1 message

Forms Response Receipts <forms-receipts-noreply@google.com>
To: yuvrajarya53@gmail.com

Mon, 13 Apr, 2026 at 12:57

Google Forms

Thanks for filling out this form: DBMS Quiz 3

Here's what was received.

DBMS Quiz 3

Your email (yuvrajarya53@gmail.com) was recorded when you submitted this form.

Name *

Raj Aryan

Enrollment No. *

2510JC032

QUESTIONS

Attempt all questions

Q. Which of the following join is used to get all the tuples of relation X and Y with Null values of all corresponding missing values? *

Consider the following relations X (S, Si, C) and Y (S, P, D).

X

S	Si	C
J	1	M
B	2	N
R	3	H
T	4	G

Y

S	P	D
J	S ₁	CA
B	P ₁	AB
R	D ₁	DC
A	H ₁	MD

- ☐ Left Outer Join
- ☐ Right Outer Join
- ☐ Natural Join
- ☒ Full Outer Join

Q. Identify the correct operation which produces R3 based on two relations R1 and R2.

Choose the correct options from the options below:

*

R1

Name	Class
a1	9
b1	5
c1	6
d1	11

R2

Name	Class
b1	5
f1	8

R3

Name	Class
b1	5

- ☐ R1 ∪ R2
- ☐ R1 × R2

☐ R1-(R1-R2)

☒ R1-R2

Q. Given the following STUDENT-COURSE scheme: STUDENT (Rollno, Name, Courseno) COURSE (Courseno, Coursename, Capacity), where Rollno is the primary key of relation STUDENT and Courseno is the primary key of relation COURSE. Attribute Coursename of COURSE takes unique values only. The number of records in COURSE and STUDENT tables are 3 and 5 respectively. Following relational algebra query is executed: **R=STUDENT X COURSE**
Match List I with List II in context to the above problem statement.

List I

- A. Degree of table R
- B. Cardinality of table R
- C. Foreign key of relation STUDENT
- D. Foreign key of relation COURSE

List II

- I. 15
- II. NIL
- III. 6
- IV. Course_no

Choose the correct answer from the options given below:

*

☐ A-I, B-III, C-IV, D-II

☐ A-III, B-I, C-II, D-IV

☐ A-I, B-III, C-II, D-IV

☒ A-III, B-I, C-IV, D-II.

Q. Given two tables:

EMPLOYEE (EID, ENAME, DEPTNO)

DEPARTMENT (DEPTNO, DEPTNAME)

Find the most appropriate statement of the given query:

Select count () 'total' from EMPLOYEE
where DEPTNO IN (D1, D2)
group by DEPTNO*

having count () > 5*

*

- ☐ Total number of employees in each department D1 and D2.
- ☒ Total number of employees of department D1 and D2 if their total is >5.
- ☐ Display total number of employees in both departments D1 and D2.
- ☐ The output of the query must have at least two rows.

Q. Given two tables R1(x, y) and R2(y, z) with 50 and 30 number of tuples respectively. Find maximum number of tuples in the output of natural join between tables R1 and R2? *

- ☐ 30
- ☐ 20
- ☐ 50
- ☒ 1500

Q. Let E1 and E2 be two entities in an E/R diagram with simple single-valued attributes. R1 and R2 are two relationships between E1 and E2, where R1 is one-to-many and R2 is many-to-many. R1 and R2 do not have any attributes of their own. What is the minimum number of tables required to represent this situation in the relational model? *

- ☐ 2
- ☒ 3
- ☐ 4
- ☐ 5

Q. The relation schema given below is used to store information about the employees of a company, where empId is the key and deptId indicates the department to which the employee is assigned. Each employee is assigned to exactly one department.

emp(empId, name, gender, salary, deptId)

Consider the following SQL query:

```
select deptId, count(*) from emp  
where  
gender="female" and salary > (select avg (salary)from emp) group by deptId;
```

The above query gives, for each department in the company, the number of female employees whose salary is greater than the average salary of

*

- ☐ employees in the department
- ☒ employees in the company
- ☐ female employees in the department
- ☐ female employees in the company

Q. Which SQL JOIN type returns all rows when there is a match in either the left or right table? *

- ☐ INNER JOIN
- ☐ LEFT JOIN
- ☐ RIGHT JOIN
- ☒ FULL JOIN

Q. The SQL query:

```
SELECT columns  
FROM TableA  
RIGHT OUTER JOIN TableB  
ON A.columnName = B.columnName  
WHERE A.columnName IS NULL
```

returns the following:

*

- ☒ All rows in Table B, which meets equality condition above and, none from Table A which meets the condition.
- ☐ All rows in Table A, which meets equality condition above and none from Table B, which meets the condition.

- ☐ All rows in Table B, which meets the equality condition
- ☐ All rows in Table A, which meets the equality condition

Q. Consider a relational database containing the schemas shown in the tables . The primary key of each table is indicated by underlining the constituent fields.

```
SELECT s.sno, s.sname
FROM Suppliers s, Catalogue c
WHERE s.sno=c.sno AND
cost > (SELECT AVG (cost)
FROM Catalogue
WHERE pno = 'P4'
GROUP BY pno) ;
```

The number of rows returned by the above SQL query is

*

Catalogue		
<u>sno</u>	<u>pno</u>	cost
S1	P1	150
S1	P2	50
S1	P3	100
S2	P4	200
S2	P5	250
S3	P1	250
S3	P2	150
S3	P5	300
S3	P4	250

Suppliers		
<u>sno</u>	sname	location
S1	M/s Royal furniture	Delhi
S2	M/s Balaji furniture	Bangalore
S3	M/s Premium furniture	Chennai

Parts		
<u>pno</u>	pname	part_spec
P1	Table	Wood
P2	Chair	Wood
P3	Table	Steel
P4	Almirah	Steel
P5	Almirah	Wood

- ☐ 4
- ☐ 5
- ☐ 0
- ☒ 2

School of Computer and Systems Sciences
Jawaharlal Nehru University
New Delhi-110067
Database Management Systems (CS-407)
Mid-Semester Exam, Winter Semester – 2026

Duration: 90 Minutes

Max. Marks: 25

All questions are compulsory.

Q1 (a) Differentiate between database and database management system. Give suitable example. (2)

Q1 (b). What do you mean by composite attributes and multivalued attributes. How do we represent them in ER diagrams and in relational data model? (3)

Q2 (a). What is referential integrity constraint and how it is being enforced in DBMS? (1)

Q2 (b). A relation R (P, Q, R, S) with a set of Functional Dependency $F = (PQ \rightarrow R, R \rightarrow S, S \rightarrow P)$ is decomposed into $R_1 (P, Q, R)$ and $R_2(R, S)$. Find out whether the decomposition is lossy/ lossless and dependency preserving or not. (4)

Q3 (a). Define primary key, super key candidate key and foreign key with example. (2)

Q3 (b). Consider the relation schema $R = (A, B, C, D, E, F, G, H, I, J)$ and the set of functional dependencies-

$AB \rightarrow C, B \rightarrow EF, AD \rightarrow GH, G \rightarrow I, H \rightarrow J$. Determine candidate keys for R. (3)

Q4 (a). Consider the relation $R(A, B, C, D, E, F, G, H)$ with following functional dependencies

$AB \rightarrow E, A \rightarrow EF, B \rightarrow F, F \rightarrow GH$

Consider the decomposition of R into constituent relations according to D_1 and D_2 decomposition schemes

$D_1 : R_1[(A, B, C, D), (A, D, E), (B, F), (F, G, H)]$

$D_2: R_2 [(A, B, C), (D, E), (B, F), (F, G, H)]$ (3)

Q4 (b). Given two sets of FDs for a relational variable R {A, B, C, D, E}, Are they equivalent?

1. $F_1: A \rightarrow B, AB \rightarrow C, D \rightarrow AC, D \rightarrow E$

2. $F_2: A \rightarrow BC, D \rightarrow AE$. (2)

Q5 Construct an E-R diagram for a car-insurance company whose customers own one or more cars each. Each car has associated with it zero to any number of recorded accidents. Date and location of each accident occurred is reported and is identified by report-number. The insurance company maintains name and address of each insured. The damage to the car met with an accident is also recorded in the database. Identify the entities, relationships in the given problem statement and construct an ER diagram. (5)

School of Computer and Systems Sciences
Jawaharlal Nehru University
End-Semester Examination, (Winter Semester 2026)

Database Management Systems (CS-407)

Q1. Consider the following relational schema (10)

Sailors (SailorID, SailorName, Rating, Age)

Reserves (SailorId, BoatId, Day)

Boats (BoatId, BoatName, Color)

Write the SQL expressions for the following queries

- i) Find the names of sailors who have reserved a red or a green boat
- ii) Find the names of sailors who have reserved at least two boats.
- iii) Find the names of sailors who have reserved all boats.
- iv) Find the names of boats which are reserved by sailors having age greater than 30
- v) Find the names of sailors having a rating greater than 4 and who have reserved a boat on Tuesday.

Q2 a) Consider the relation $R = \{P, Q, R, S, T, U\}$ and the set of functional dependencies $F = \{P \rightarrow RS, Q \rightarrow TU, R \rightarrow S, T \rightarrow U\}$. Decompose R into 2NF and 3NF relations. (5,2,3)

b) Compute the canonical cover of $F = \{A \rightarrow BC, B \rightarrow AC, C \rightarrow AB\}$.

c) What is deadlock and explain Wait-and-Die and Wound-and-Wait deadlock prevention techniques.

Q3 Consider the following relational schema (10)

Employee(EmpID, EmpName, Street, City)

Works(EmpID, CompID, Salary)

Company(CompID, CompName, City)

Write the relational algebra expressions for the following queries:

- i) Find the names of employees whose salary is more than the average salary of all employees of their company.
- ii) Find the name of the employee with second highest salary in the company.
- iii) Find the names of highest paid employees in each company.
- iv) Delete the companies that have the smallest average salary.
- v) Find the name of the companies that have highest average salary.

Q4 a) Consider the schedule S and determine whether the schedule S can be executed under Time-stamp based protocol? Consider the timestamp of $T1$ and $T2$ as 10:00 pm and 10:15 am respectively and system clock is initially set to zero.

$T1$: Read (A) ; $T1$: $A = A + 50$; $T1$: Read (B) ; $T2$: Read (A) ; $T2$: Read (B) ; $T2$: $A = A * 2$; $T2$: $B = B + 30$; $T2$: Write (A) ; $T2$: Write (B) ; $T1$: $B = B * 3$; $T1$: Write (A) ; $T1$: Write (B) ; (4)

b) Differentiate between strict and conservative 2PL. (2)

Q4 a) Consider a relations $R(L, M, N, O, P, Q)$ with MVDs $L \twoheadrightarrow M$ and $NO \twoheadrightarrow PQ$. Determine whether the relation R is in 4NF or not and decompose it into 4NF (if not). (3,4)

b) Decompose the relation $R = \{A, B, C, D, E, F\}$ with the set of functional dependencies $F = \{A \rightarrow BC, F \rightarrow DE\}$ and join dependency $JD[ABC, ADF, DEF]$ into relations in 5NF.

Q5 a) Consider a relation $R(A, B, C)$. Determine if the instance $R\{(a1, b1, c1), (a1, b2, c2)\}$ exhibit join dependency $JD[AB, BC, AC]$. If a tuple $(a2, b1, c2)$ is added to the instance R then how is join dependency $JD[AB, BC, AC]$ ensured? (4,3)

b) Determine the conflict equivalent schedule for the following schedule:

$T1$: $R(X)$, $T2$: $R(Y)$, $T3$: $R(Y)$, $T2$: $W(Y)$, $T1$: $W(X)$, $T3$: $W(X)$, $T2$: $R(X)$, $T2$: $W(X)$